

Formalizing quantum-fluid structure in Lean 4:

a machine-checked library, what the formalization caught,
and what it could not

Xavier Callens

SocrateAI-Scientific-QuantumFluids

<https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids>

September 2026

Abstract

We present an open Lean 4/Mathlib library of 109 machine-checked theorems on the mathematical structure of quantum fluids: correctness of the phase-winding algorithm by which every Gross–Pitaevskii code locates vortices, quantization of circulation in the discrete form in which it is actually computed, sign and conservation structure of the Galerkin-truncated Gross–Pitaevskii nonlinearity on an arbitrary mode set, the Madelung decomposition, and a bridge that restates these objects inside the vocabulary of OpenAI’s Lean formalization of the Navier–Stokes problem, imported as a real dependency. Every theorem carries a generated axiom audit and is independently re-checked against two proof kernels.

This is not a physics-discovery paper, and we say why. The project began as one. Two results we had regarded as findings were withdrawn after a literature check: a cubic invariant of a complexified dyadic shell model is a special case of the Hamiltonian structure of Vladimirova, Shavit and Falkovich and of L’vov, Podivilov and Procaccia; and a “dual length” bound on the Bogoliubov dispersion is dimensionally forced and weaker than Onsager’s inequality where it is not vacuous. We report these retractions in full, because they are the clearest data we have on the actual subject of the paper: *what formal verification does and does not buy in physics*. It caught a false lemma we were about to state, an inequality our prose had turned into an equality, a constant that made a bound useless in the limit that mattered, eleven theorems that had silently escaped audit, and a lemma whose `private` modifier had exempted it from independent checking. It could not tell us that a correct theorem was already known, nor that a hypothesis was physically naive; a literature search and a simulation did that. We give the library to the community as verification infrastructure, with the open formalization targets we consider most valuable.

1 Introduction

Proof assistants are now used at scale on fluid equations: OpenAI’s announced constructions for the Navier–Stokes and Euler equations ship with a Lean 4 development of several thousand files [1]. Quantum fluids—superfluid helium, atomic Bose–Einstein condensates—have, to our knowledge, no comparable formal footprint, although they are in one respect better suited to it: circulation is quantized, the healing length is fixed by constants, and the central algorithms of the field (vortex detection by phase winding, Fourier–Galerkin truncation) are exactly discrete, so their correctness statements are finite and provable rather than asymptotic.

This paper contributes a library, and a candid account of building it. The account matters as much as the library. Over roughly six weeks the project produced, alongside correct proofs, a sequence of claims that did not survive: two that a literature check identified as rediscoveries or dimensionally forced (§5), and a topological hypothesis that a well-resolved simulation contradicts (§6). Each failure was caught by a mechanism that was in place *before* the claim was tested—a pre-registered control, an external review, a literature gate. We think the distribution of which

mechanism caught which error is informative for anyone using proof assistants in the physical sciences, and §4–5 are organised around it.

Stance. No novelty is claimed for any physical statement in this paper. The contribution is (i) the machine-checked library and its verification tooling, (ii) one correctness result for a standard algorithm that we have not found stated elsewhere with its failure case (§3.1), and (iii) the error taxonomy.

2 The library

Build. Lean 4.34.0-rc2 with Mathlib at tag v4.34.0-rc2. A single `import QuantumFluids` brings in fourteen modules (Table 1); the full build is 8792 jobs, zero errors, zero sorry in any default target.

module	thms	content
VortexWinding	7	loop quantization of the principal-branch phase sum; edge cancellation and its exact failure case
QuantizedCirculation	6	$\Gamma = q\kappa$, $\kappa = h/m$; no fraction of a quantum; the quantum is attained
GPGalerkin	13	truncated Gross–Pitaevskii on any finite mode set: positivity of the interaction sum, mass and energy algebra, Hamiltonian gradient identity
HeliumKinematics	15	identities a neutron-scattering analysis of ^4He rests on: three-phonon decay open <i>iff</i> the dispersion is anomalous, two-roton momentum range, calibration-invariance of a comparison with $2\Delta_R$, time-of-flight equivalences
BoseIntegral	5	$\int_0^\infty t^{s-1}/(e^t - 1) dt = \Gamma(s)\zeta(s)$ via Mathlib’s Mellin machinery; $\int_0^\infty t^3/(e^t - 1) dt = \pi^4/15$; the Debye T^3 coefficient
MadelungSplit	6	$ \nabla\psi ^2 = \nabla a ^2 + a^2 \nabla S ^2$
MadelungNSE	2	the Madelung velocity in the vocabulary of [1]; $\nabla \cdot \nabla\varphi = \Delta\varphi$ in their definitions
DualLength	13	algebra of $\varepsilon^2/(\hbar^2 c^2 k^3)$ for the Bogoliubov dispersion (significance withdrawn, §5)
QuantumFluidsShell	12	complexified dyadic shell model: conservation, invariant real subspace, trace identities
ShellHamiltonian, SigmaRule	15	its cubic invariant and cutoff-uniform bounds (novelty withdrawn, §5)
Duality, RipsFloor	15	auxiliary inequalities; the 1-skeleton lemma used by the topological analysis

Table 1: Modules. Axiom footprint of all 109 theorems: {proptext, Classical.choice, Quot.sound}.

Verification stack. Three layers, each added after a specific failure.

1. *Generated axiom audit.* A script regenerates the `#print axioms` block of every file from its declarations and has a `--check` mode. It replaced hand-kept lists after an external review found those had drifted (§4, item 5).
2. *Comparator* [2]. For each module a challenge file—definitions verbatim, proofs replaced by `sorry`—is generated from the solution, and Comparator checks statement identity, an axiom whitelist, and acceptance by both the Lean kernel and the independent `nanoda` kernel, inside

a sandbox. Because challenges are generated from solutions, statement identity holds by construction; the independent content is the second kernel and the whitelist. Whether a statement is the *right* statement remains a human audit, and §4 shows why that caveat is not a formality.

3. *Negative controls.* For each headline theorem a perturbed statement (a wrong constant, a dropped conjugate, a flipped inequality) is compiled and required to fail. A checker that has never been seen to reject is not yet a checker.

3 Selected results

3.1 Correctness of phase-winding vortex detection

Every Gross–Pitaevskii simulation code locates quantized vortices the same way: sum the principal-branch phase differences of ψ around a grid plaquette, divide by 2π , round. Let $\text{pd}(a, b) \in (-\pi, \pi]$ denote the principal representative of $b - a$.

Theorem 1 (`loop_sum_eq_mul`). *For any phases $\theta_0, \dots, \theta_n$ with $\theta_n = \theta_0$, $\sum_{i < n} \text{pd}(\theta_i, \theta_{i+1}) \in 2\pi\mathbb{Z}$.*

So the computed winding number is an integer by theorem, not by numerical accident. The second fact codes rely on—that an edge shared by two plaquettes contributes oppositely to each, so windings add like a discrete Stokes theorem—is usually stated as antisymmetry, $\text{pd}(a, b) = -\text{pd}(b, a)$. **That statement is false**, and attempting to prove it is how we found out:

Theorem 2 (`pdiff_add_rev`, `pdiff_add_rev_eq_two_pi_iff`). $\text{pd}(a, b) + \text{pd}(b, a) \in \{0, 2\pi\}$, and the value 2π occurs exactly when $\text{pd}(a, b) = \text{pd}(b, a) = \pi$.

At a phase step of exactly π both traversal directions return $+\pi$, cancellation between neighbouring plaquettes fails, and a spurious unit of winding can appear. The folklore criterion “resolve the field so that no phase step reaches π ” is thereby a theorem with a sharp boundary. We have not found the disjunctive form stated in the literature; it is elementary, and we claim only that it is now checked.

3.2 Quantized circulation

With superfluid velocity $u = (\hbar/m)\nabla S$, define the discrete circulation around a closed loop of sample points as $\Gamma = (\hbar/m) \sum_i \text{pd}(S_i, S_{i+1})$ and $\kappa = 2\pi\hbar/m$.

Theorem 3 (`circulation_quantized`, `abs_circulation_ge`, `circulation_quantum_attained`). $\Gamma = q\kappa$ for an integer q ; if $\Gamma \neq 0$ then $|\Gamma| \geq \kappa$; and an explicit four-point loop attains $\Gamma = \kappa$.

The attained case is included deliberately: a lower bound with no witness can be vacuous, and the witness’s closing step wraps by 2π , exercising exactly the branch behaviour the definition exists to handle. *Not proved*: that the discrete sum equals the continuum line integral. That is a sampling statement, and by the previous subsection it fails, by design, when a phase step reaches π .

3.3 Truncated Gross–Pitaevskii on an arbitrary mode set

Let G be an additive commutative group ($\mathbb{Z}^3$ in applications), $\Lambda \subset G$ finite, and $N_k = \sum_{k_1+k_3=k+k_2} \psi_{k_1} \psi_{k_2} \psi_{k_3}$ the projected cubic nonlinearity.

Theorem 4 (`pairing_eq_sum`, `pairing_nonneg`, `grad_identity`, `energy_rate_zero`). *For every finite Λ : $Q := \sum_{k \in \Lambda} \overline{\psi_k} N_k = \sum_q |A_q|^2 \geq 0$ with $A_q = \sum_{a-b=q} \psi_a \overline{\psi_b}$; $\sum_k \text{Im}(\overline{\psi_k} G_k) = 0$ for $G_k = \omega_k \psi_k + g N_k$; the polarised gradient identity $\sum_q 2\text{Re}(\overline{A_q} dA_q[\delta]) = 4\text{Re} \sum_k \overline{\delta_k} N_k$ holds; and the first variation of $E = \sum \omega_k |\psi_k|^2 + \frac{g}{2} Q$ along $\delta = -iG$ vanishes.*

The statements are uniform in the truncation: no property of Λ beyond finiteness is used, which is the form in which such identities are needed when a cutoff is to be removed. What is proved is the algebraic core; the chain-rule step identifying these variations with time derivatives along a solution is not formalised, and the module header says so.

3.4 A cross-project import

`MadelungNSE` imports `NavierStokes.ProblemStatement` from [1] as a Lake dependency pinned to a commit whose headline theorems had been independently audited. In *their* definitions of divergence and pressure gradient we prove $\nabla \cdot \nabla \varphi = \Delta \varphi$ —absent from their tree, where pressure enters only through its gradient—and hence $\nabla \cdot u = (\hbar/m)\Delta S$ for the Madelung velocity. The mathematics is elementary. The point is operational: the import became possible only when both projects were aligned on one toolchain and one Mathlib revision; before that, version skew, not missing theorems, was the entire obstacle to reuse. One convenience lemma was dropped rather than carried with a `sorry` when it did not survive a coercion change in Mathlib; nothing depended on it.

4 What the formalization caught

Each item is an error that existed in prose, code, or intent, and was surfaced by the act of formalizing or by the verification stack.

1. **A false lemma, before it was used.** Antisymmetry of the principal phase difference (§3.1). Our own extractor’s documentation asserted it.
2. **An inequality that prose had made an equality.** A design memo described a structure-factor bound as holding “exactly” under Feynman’s relation. The Lean theorem could only be stated with the Bijl–Feynman relation as an *inequality* hypothesis, $\varepsilon \leq \hbar^2 k^2 / (2mS)$, which is what it is. The formal statement was right while the accompanying prose was wrong; the discrepancy was found when the two were compared.
3. **A constant that voids a bound where it matters.** A cutoff-uniform enstrophy bound for a third-order dispersive regulator, $2D\Omega \leq H + (2E)^{3/2}$, looked relevant to a uniformity question in a sibling project. Written out for the formal statement, the bound on Ω carries $1/D$ and diverges as the regulator is removed—precisely the class of bound that project had already ruled out. We had described the result as possibly significant; the fully explicit statement is what retracted that.
4. **Hypotheses that locate the physics.** The Madelung identities require differentiability of amplitude and phase at the point. That hypothesis is not bureaucratic: it fails exactly on a vortex line, which is where the hydrodynamic picture itself fails. The formal statement makes the domain of validity part of the theorem.
5. **Audit drift.** An external reviewer rebuilding the tagged release reported 13 theorems in a file documented as having 12. Checking every file showed 76 theorems against 65 audit lines: eleven theorems had never been axiom-audited, and every count we had published conflated “theorems” with “theorems carrying an audit line”. All eleven were clean; the defect was in the reporting. The generated audit (§2) is the fix.
6. **A lemma silently exempt from independent checking.** One helper was declared `private`. A private name is mangled, so it cannot be exported to the second kernel: the run aborted rather than passing, and only then did we notice that the lemma had never been independently checked. It is now public. A visibility modifier had been acting as an unintended exemption from verification.

7. **A clone-breaking artefact.** The same review found the build directory committed as a symlink to a local disk, because an ignore pattern with a trailing slash matches directories and not symlinks. No proof assistant catches this; a second person building from the tag did.

5 What it could not catch

Two results were correct, machine-checked, and not contributions. The kernel certifies that a statement follows from its hypotheses. It is silent on whether the statement is new, and on whether it could have come out otherwise.

A rediscovery. We had found that the complexified dyadic shell model $\dot{v}_n = k_{n-1}v_{n-1}^2 - k_n\overline{v}_nv_{n+1}$ becomes Hamiltonian under $w_n = 2^{-n/2}v_n$, with a cubic invariant $H = \sum_n 2^{-n}[\omega_n|v_n|^2 + k_n \operatorname{Im}(\overline{v}_n^2 v_{n+1})]$, energy conservation as a Manley–Rowe relation, and volume preservation as a corollary; we proved it in Lean and confirmed by a nullspace search that the invariants are exactly $\{\text{mass}, \text{mass}^2, H\}$. All of this is a degenerate case of published structure: the canonical Hamiltonian form of the Sabra model under the analogous rescaling and its cubic invariant [3, 4]; the three-wave chain with graded phase symmetry and Manley–Rowe families [5]; second-harmonic generation as a minimal turbulence model [6], whose title is the framing we had taken for an insight; and the two-mode invariant itself [7]. Phase-space volume preservation is a stated *design criterion* for shell models [8]. The dyadic case is, if anything, poorer: $2^a + 2^b = 2^c$ forces $a = b$, so its Manley–Rowe family has one member where the Fibonacci lattice has two.

A dimensionally forced bound. For the Bogoliubov dispersion, which is the sum of squares of its phonon and free-particle branches [9], $\ell(k) = \varepsilon^2/(\hbar^2 c^2 k^3)$ equals $1/k + k/k_*^2$ with $k_* = 2mc/\hbar$, is invariant under $k \mapsto k_*^2/k$, and is bounded below by $\sqrt{2}\xi$. We proved this, and measured that superfluid ^4He violates the bound by factors of 21 to 51 across seven pressures using the published dispersion of [10]. But a theory with one length and one velocity admits no other outcome: any such length with an interior minimum has it near $1/\xi$ with value near ξ . The implied structure-factor bound $S(k) \leq \sqrt{k/2k_*}$ is weaker than Onsager’s inequality $S(k) \leq \hbar k/2mc$ [11, 12] for $k < k_*/2$ and vacuous beyond $2k_*$. And since $\ell = v_L^2/(c^2 k)$ with $v_L = \varepsilon/\hbar k$, any fluid with a roton violates the bound: the measurement confirms that ^4He has a roton. A proposed test on a dilute condensate [13] was therefore not run; its outcome is forced.

The lesson we draw. Formal verification moved our error rate on *internal* questions—is this statement true, are its hypotheses complete, is this constant where we think it is—close to zero. It did nothing for the *external* questions: is it known, is it forced, does it matter. Both retractions above were obtained by a literature search we had listed as a blocking gate and then ran late. In a field where the mathematics is a century deep, that gate should come first; a proof assistant makes it cheaper to be rigorously unoriginal.

6 Empirical companion work

Alongside the library we pursued a topological test: quantized circulation with a finite core suggests a minimum vortex separation near ξ , which would appear as a floor in the H_0 persistence of the vortex configuration (computed with GUDHI [14]; the only formal ingredient is the 1-skeleton lemma of RipsFloor). The sequence of outcomes is instructive mainly about controls.

1. On a published 256^3 Gross–Pitaevskii snapshot [15], proximity-based line segmentation gave a floor of 1.49ξ against a null of 0.10ξ . A sweep of the segmentation threshold showed the

floor tracking the threshold (ratio 1.00–1.24 over a fourfold change): the statistic measured its own parameter. Withdrawn.

2. Threshold-free tracing by cube adjacency (no tunable parameter; zero ambiguous cubes in 46 762) gave 0.94ξ , six times the null—but a second pre-registered criterion, the fraction of separations below ξ , was indistinguishable from the null. Reported as a split verdict; that dataset resolves ξ by only 1.5 grid cells.
3. A survey found no public dilute-condensate dataset resolving ξ by five cells or more (best: 2.3), so we wrote a split-step solver, validated by energy convergence, core formation at the correct size, and dipole annihilation, and ran at $\xi/\Delta x = 8$. Two properties of the method had to be learned and are now pinned as tests: imaginary-time relaxation first forms vortex cores and then *destroys* the vortices, and the timestep must respect $\Delta t k_{\max}^2/2 \ll 1$ (a run fifty times over the bound is detectably wrong). At $t = 20$ the minimum separation is 0.28ξ ($\Delta t = 0.01$) and 0.38ξ ($\Delta t = 0.005$), inside the matched Poisson band (0.09 – 0.78ξ) and far below ξ ; and the fraction of separations below ξ , 0.07 – 0.08 , lies *above* the null’s 95th percentile (0.04). Vortices sit *closer* than random placement, not farther. Bulk statistics are converged in the timestep (count 3%, mean separation 2%); the minimum, an extreme statistic of a chaotic system, is not (34%), and the energy drift (2.4%) still exceeds the solver’s own strict criterion, so the refutation rests on the verdict being identical at both timesteps rather than on any single number. **By the pre-registered criteria the hypothesis is refuted.**

The hypothesis was physically naive. Opposite-sign vortices attract and annihilate, a process that passes continuously through separations below ξ ; a snapshot of a live configuration therefore contains pairs in mid-approach. Quantization constrains circulation, not distance.

7 Applying the library to published experimental work

The clearest use we have found for a library of this kind is not to generate physics but to check derivations that an experimental paper states without proof. We took as subject the two publications of Godfrin and co-workers in 2021–2026: the pressure-dependent dispersion of superfluid ^4He from inelastic neutron scattering [10] and its companion review.

`HeliumKinematics` formalises relations that paper’s analysis leans on: three-phonon decay is kinematically open *iff* the dispersion is anomalous, with excess energy exactly $3ca k_1 k_2 (k_1 + k_2)$; two rotons carry total momentum in $[0, 2k_R]$, attained by parallel and anti-parallel pairs, which is the review’s statement that the Pitaevskii plateau “ends at $2k_\Delta$ ” and “can in principle extend to zero wave number”; every time-of-flight energy is proportional to the incident energy, so *any* comparison of $\varepsilon(k)$ with $2\Delta_R$ is invariant under recalibration of the roton gap; and Eq. (7) of [10] is $dP/d\rho$ of its Eq. (6).

Two consequences are worth reporting. First, with the published low- k coefficients each closing condition factors as ck^n times a *quadratic*, so the thresholds are explicit: soft emission closes at 0.404Å^{-1} , the symmetric split at 0.455, the phase velocity reaches c at 0.566. These are three different numbers, often quoted as one k_c , and the symmetric split is the last two-phonon channel to close; the measured table alone, with no fitted series, gives 0.453. Second, `BoseIntegral` verifies the leading coefficient of that paper’s specific-heat series — the series for which it notes that earlier published versions “contain errors” — from $\int_0^\infty t^3/(e^t - 1) dt = \pi^4/15$ through to $A = 2\pi^2 k_B^4 V/(15c^3 \hbar^3) = 0.0831\text{ J}/(\text{mol K}^4)$, matching the printed value. Mathlib had $\zeta(4)$ and the Gamma integral but not the Bose integral.

We also record two apparent misprints in the preprint, demonstrated numerically against the paper’s own tables rather than asserted, in `docs/FOR.GODFRIN.md`. That note has not been sent; it is a draft for the repository owner.

8 A proposal the literature gate stopped

§5 concluded that the search for prior art should come first. Tested on the next proposal, it did. We intended to apply 0-dimensional sublevel persistent homology to the measured dispersion: the roton as a homology class born at Δ_R and dying at the maxon Δ_M , with small-persistence pairs as instrumental noise. For a function on an interval this is exactly topographic prominence — as the founders of the subject state themselves — and on the published table the numbers agree to the digit: prominence 0.4501 meV against $\Delta_M - \Delta_R = 0.4501$ meV. The persistence of the roton is a difference of two numbers already tabulated at every pressure. Sublevel persistence of a dispersion with a noise threshold is in any case already published for photonic band structures, and persistent homology of a Gross–Pitaevskii gas with quantized vortices is published too — which our own refuted vortex work of §6 should have cited. Nothing was built. The episode cost a literature search rather than a release.

9 Using and extending the library

Reuse. The modules marked in §2 are self-contained. `VortexWinding` and `QuantizedCirculation` depend only on Mathlib and apply to any code using plaquette phase winding. `GPGalerkin` is stated over an arbitrary additive group and finite mode set.

Open targets, in the order we consider most valuable:

1. the sampling theorem linking the discrete loop sum to the continuum circulation for a C^1 phase with step bound below π —this would turn resolution criteria for vortex detection from folklore into theorem;
2. the oriented sum of plaquette windings over the faces of a grid cube vanishes (away from the π case)—the discrete statement that vortex lines do not end, which is what threshold-free line tracing silently assumes;
3. the chain-rule step for `GPGalerkin`, upgrading algebraic rates to conservation along solutions of the truncated system;
4. the Gross–Pitaevskii residual in the vocabulary of [1], parallel to their Navier–Stokes residual, with the Madelung correspondence as a theorem between them;
5. the remaining coefficients C, D, E, K, L of [10] Eq. (22), which need the same Bose integral at $s = 6 \dots 10$ together with a Lagrange inversion — the target with the most direct value to an experiment, since the paper reports a disagreement with two earlier published series.

10 Conclusion

We set out to find new structure in quantum fluids with a proof assistant and found, after checking, that the structure we had was known. What remains is a library whose statements are exactly as strong as they say, a small correctness result for a ubiquitous algorithm, and a record in which every error is attributed to the mechanism that caught it. The record suggests a division of labour: the kernel for truth, controls for measurement, a second person for the build, and the library for novelty. We offer the first of these, with the other three as its documented context.

Availability. Source, claim ledger with all retractions (`RETRACTIONS.md`), pre-registrations and run outputs: <https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids>. The earlier report on the shell-model measurement programme remains in the repository at tag `v1.1.0`.

Acknowledgements. An external reviewer working from the LeanMaster project rebuilt the tagged release independently and reported the two defects of §4, items 5 and 7. The ^4He dispersion data are those of Godfrin et al. [10]; the turbulence snapshot is that of Polanco, Müller and Krstulovic [15].

References

- [1] OpenAI. Finite time blowup for Navier–Stokes and Euler equations: Lean 4 formalization. <https://github.com/openai/NavierStokesAndEuler>, 2026. commit 8937a8f.
- [2] Lean FRO. Comparator: a trustworthy judge for Lean proofs. <https://github.com/leanprover/comparator>, 2026.
- [3] V. S. L’vov, E. Podivilov, and I. Procaccia. Hamiltonian structure of the Sabra shell model of turbulence: exact calculation of an anomalous scaling exponent. *Europhys. Lett.*, 46:609, 1999. arXiv:chao-dyn/9804036.
- [4] P. D. Ditlevsen. Symmetries, invariants and cascades in a shell model of turbulence. *Phys. Rev. E*, 62:484, 2000. arXiv:chao-dyn/9811004.
- [5] N. Vladimirova, M. Shavit, and G. Falkovich. Fibonacci turbulence. *Phys. Rev. X*, 11:021063, 2021. arXiv:2101.10418.
- [6] N. Vladimirova, M. Shavit, S. Belan, and G. Falkovich. Second harmonic generation as a minimal model of turbulence. *Phys. Rev. E*, 104:014129, 2021. arXiv:2103.15468.
- [7] J. A. Armstrong, N. Bloembergen, J. Ducuing, and P. S. Pershan. Interactions between light waves in a nonlinear dielectric. *Phys. Rev.*, 127:1918, 1962.
- [8] L. Biferale. Shell models of energy cascade in turbulence. *Annu. Rev. Fluid Mech.*, 35:441–468, 2003.
- [9] F. Dalfovo, S. Giorgini, L. P. Pitaevskii, and S. Stringari. Theory of Bose–Einstein condensation in trapped gases. *Rev. Mod. Phys.*, 71:463, 1999. arXiv:cond-mat/9806038.
- [10] H. Godfrin, K. Beauvois, A. Sultan, E. Krotscheck, J. Dawidowski, B. Fåk, and J. Ollivier. Dispersion relation of Landau elementary excitations and thermodynamic properties of superfluid ^4He . *Phys. Rev. B*, 103:104516, 2021.
- [11] W. F. Wreszinski and M. A. da Silva Jr. Onsager’s inequality, the Landau–Feynman ansatz and superfluidity. *J. Phys. A*, 38:6293, 2005. arXiv:cond-mat/0411640.
- [12] S. Stringari. Sum rules and Bose–Einstein condensation, 1993. arXiv:cond-mat/9311024.
- [13] J. Steinhauer, R. Ozeri, N. Katz, and N. Davidson. Excitation spectrum of a Bose–Einstein condensate. *Phys. Rev. Lett.*, 88:120407, 2002.
- [14] The GUDHI Project. GUDHI user and reference manual. <https://gudhi.inria.fr>, 2025. version 3.13.
- [15] J. I. Polanco, N. P. Müller, and G. Krstulovic. Sample generalised Gross–Pitaevskii data for circulation statistics. Zenodo, 2021.